Engine Model(s): PT6A

75-30-00

COMPRESSOR BLEED VALVE - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, which is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC05-007	Fluid, Check, Leak
PWC09-003	Compound, Silicone Sealing
PWC09-020	Compound, Lubricating
PWC11-014	Alcohol, Isopropyl
PWC11-021C	Compound, Cold Carbon Removing
PWC11-027	Solvent, Petroleum

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

The fixtures, equipment and supplier tools listed below are used in the following procedures.

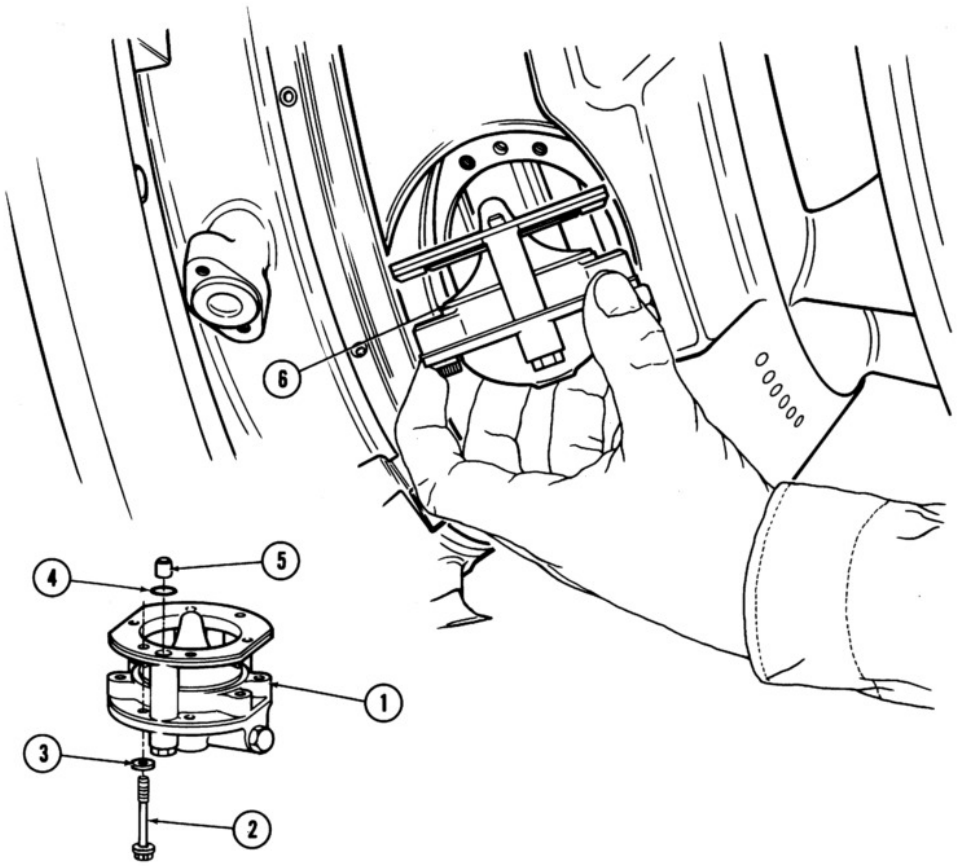
Name	Remarks
Air Pressure Regulator	100 psi input, 0 to 100 psi output
Shut-off Valve (Air Line)	
Pressure Gage	0 to 75 psi, accuracy ± 5 %

5. Removal/Installation

A. Removal of Compressor Bleed Valve (Ref. Fig. 201)

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Figure 201 Removal/Installation of Compressor Bleed Valve (Typical)



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1. Compressor Bleed Valve
2. Bolt
3. Washer
4. Preformed Packing
5. Spring Pin
6. Valve Piston

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CAUTION: KEEP THE COMPRESSOR INLET AREA CLEAR OF ALL FOREIGN OBJECTS THAT COULD BE INGESTED BY THE ENGINE AND CAUSE SERIOUS INTERNAL DAMAGE.

NOTE: In order to gain access to the bleed valve, remove full width of air inlet screen (Pre-SB1242) (Ref. 72-20-00).

(1) Remove the four bolts (2) and washers (3) securing the bleed valve (1) to the gas generator case.

(2) Withdraw the bleed valve from the gas generator case together with the spring pin (5).

(3) Remove and discard the preformed packing (4) from the spring pin.

(4) Check that P3 input line to bleed valve is free from obstructions.

(a) Apply clean, dry compressed air at P3 port of bleed valve mounting boss in gas generator case.

NOTE: There must be free air flow.

B. Installation of Compressor Bleed Valve (Ref. Fig. 201)

CAUTION: KEEP THE COMPRESSOR INLET AREA CLEAR OF ALL FOREIGN OBJECTS THAT COULD BE INGESTED BY THE ENGINE AND CAUSE SERIOUS INTERNAL DAMAGE.

NOTE: Manually modulate the BOV piston to ensure easy movement of the piston before use.

(1) Fit a new preformed packing (4) over the spring pin (5) and insert the pin into the base of the bleed valve (1).

(2) Make sure the piston (6) is free to travel. Locate the bleed valve on the mounting pad of the gas generator case, making sure the spring pin (5) enters the alignment hole in the pad and dowel pin in the base of valve is correctly seated.

(3) Secure the valve with four bolts (2) and washers (3). Tighten the bolts 16 to 20 lb.in., and secure with lockwire.

(4) Install any parts in the air inlet screen area that were removed in order to gain access to the valve (Ref. 72-20-00).

6. Cleaning/Painting

A. Cleaning

(1) Clean all the disassembled parts (Ref. Para. 8.) of the compressor bleed valve:

(a) Clean all the metallic parts in petroleum solvent (PWC11-027) and allow to drain. Dry with clean, dry compressed air.

(b) Clean the bore of the spacer sleeve with a swab moistened with isopropyl alcohol (PWC11-014). Since the spacer is a graphite-filled polyamide-resin sleeve, all dark traces cannot be removed.

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WARNING: CARBON REMOVING COMPOUND (PWC11-021C) IS TOXIC AND CAUSTIC. IT IS QUICKLY ABSORBED THROUGH THE SKIN CAUSING SEVERE BURNS AND CAN BE FATAL IF SWALLOWED. ALL PRECAUTIONS MUST BE TAKEN TO PREVENT CONTACT WITH SKIN AND EYES AND DO NOT BREATH VAPORS. IF YOUR EYE OR SKIN ACCIDENTALLY CONTACT WITH COMPOUND, WASH THOROUGHLY WITH WATER FOR AT LEAST 15 MINUTES. IF YOU ACCIDENTALLY SWALLOWED, GET MEDICAL AID IMMEDIATELY.

(c) Soak the primary slotted metering plug and final metering plug in cold carbon removing compound (PWC11-021C) for two hours at room temperature. Do not intermingle parts from different valves since the orifices are calibrated as a set to the assembly from which they were removed. Dry the exterior of the plugs with lint-free cloth and blow out the orifice using clean, dry compressed air.

(d) If the metering plugs remain contaminated or blocked after soaking, use a wooden dowel to remove carbon deposits and repeat the cleaning procedure.

NOTE: If the plugs cannot be cleaned satisfactorily, return the complete valve assembly to an approved overhaul facility.

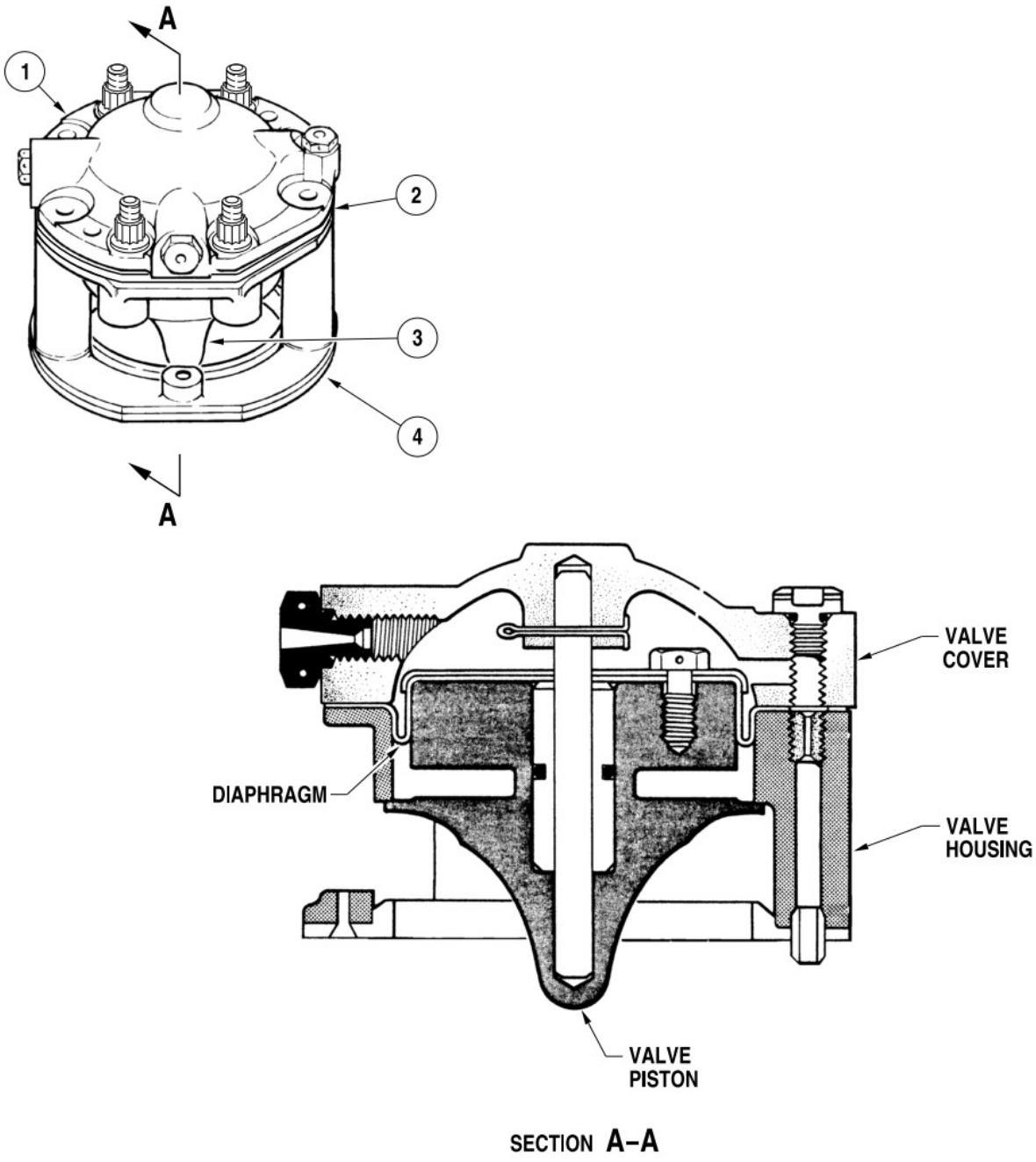
7. Inspection/Check

NOTE: Return the compressor bleed valve to an approved overhaul facility for the replacement of components, except where repair is permitted (Ref. Para. 8.).

A. BOV Assembly (Ref. Fig. 202)

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Figure 202 Inspection of Compressor Bleed Valve



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1. Valve Cover
2. Diaphragm
3. Valve Piston
4. Valve Housing

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(1) Examine for cracks, dents, nicks and other damage.

(2) Examine sealing and mating surfaces for scratches, galling, scoring and other surface damage, particularly those parts affected by travel of piston and rolling diaphragm (viewable through exit ports).

(3) Examine bleed valve seat for surface damage at piston contact face.

(4) Check for integrity of the diaphragm.

(5) Do a leak check (Ref. Adjustment/Test).

B. BOV Components (Ref. Fig. 202)

(1) When a compressor bleed valve has been disassembled and cleaned (Ref. Paras. 6. and 8.), inspect the components:

(a) Examine the valve housing (4) and cover (1) for cracks, dents, nicks and other damage.

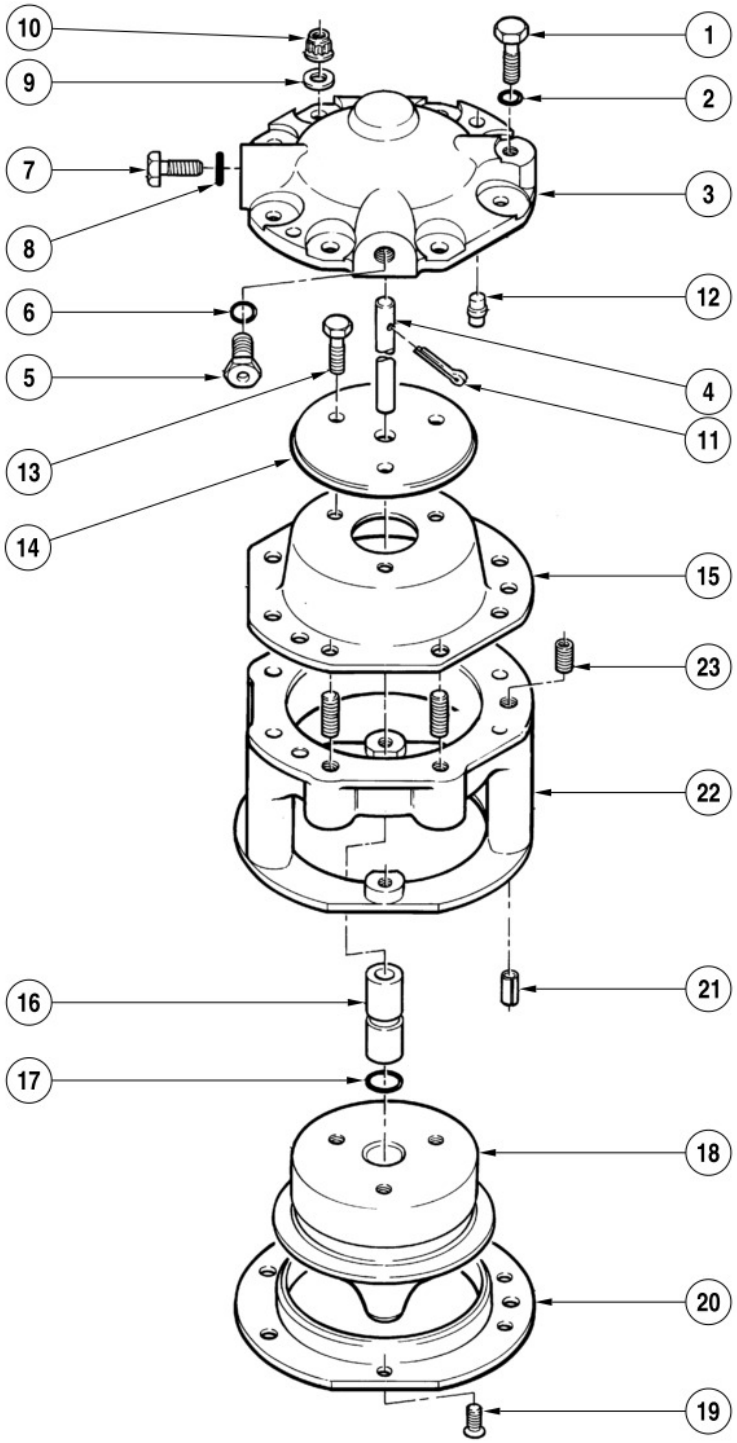
(b) Inspect the sealing, sliding and mating surfaces for scratches, galling, scoring and other surface damage, particularly those parts affected by the travel of the piston and rolling diaphragm.

8. Approved Repairs

A. Disassembly of Metering Plugs (Ref. Fig. 203)

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Figure 203 Disassembly/Assembly of Compressor Bleed Valve



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1. Blanking Plug
2. Preformed Packing
3. Valve Cover
4. Guide Pin
5. Metering Plug
6. Preformed Packing
7. Blanking Plug
8. Preformed Packing
9. Washer
10. Self Locking Nut
11. Cotter Pin
12. Shoulder Pin
13. Bolt
14. Retaining Plate
15. Diaphragm
16. Sleeve Spacer
17. Preformed Packing
18. Piston
19. Screw
20. Seat
21. Spring Pin
22. Valve Housing
23. Metering Plug - Primary Orifice

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NOTE: Metering plugs should only be removed for inspection and/or cleaning requirements. Plugs must never be intermixed with parts from other valves or new parts fitted in the field.

(1) Remove final metering plug (5) from valve cover (3). Remove preformed packing from plug.

(2) Remove blanking plug (1) from cover (3) and primary metering plug (23) from valve housing (22), with an appropriate screwdriver.

B. Assembly of Metering Plugs (Ref. Fig. 203)

(1) After cleaning and inspection, assemble metering plugs:

(a) Install metering plug (23) in P3 cored passage of valve housing (22). Torque plug 10 to 20 lb.in.

(b) Install preformed packing on blanking plug (1), and fit plug in cover (3), torque 15 to 20 lb.in.

(c) Install new preformed packing on final metering plug (5) and install plug in valve cover (3). Do not torque plug at this time.

(d) Do adjustment/test.

NOTE: If testing is done immediately after Repair procedure, operators may choose to leave blanking plug (1), and metering plug (23) installed. Make sure that no contamination enters open ports in valve cover (3).

C. Replacement of Bleed Valve Diaphragm (Ref. Fig. 203)

(1) Remove four nuts (10) and washers (9) that attach valve cover (3) to valve housing (22). Remove cover.

(2) Remove three bolts (13) that attach diaphragm retaining plate (14) and diaphragm (15) to piston (18). Remove retaining plate and diaphragm. Refer to Para. 7. for inspection of components.

(3) Lightly smear top of piston (18), mating face of housing (22) and both flange mating faces of diaphragm with compound (PWC09-003) paying particular attention to area around metering plug (23). Make sure metering plug is dry and cavity is free of compound.

(4) Install new diaphragm (15) with overall fabric surface and identification ink stamp against piston, and diaphragm retaining plate (14) on valve piston, making sure skirt of diaphragm is correctly positioned on studs of valve housing and offset hole is aligned with P3 cored passage. Attach diaphragm and retaining plate with three bolts (13).

(5) Push piston through towards seat end of valve housing until diaphragm skirt is correctly seated on studs.

(6) Holding diaphragm skirt securely on flange, make sure diaphragm rolls correctly between piston and housing when piston is moved up and down in housing.

CAUTION: RETAINER MUST BE EXACTLY CENTERED TO MAKE SURE THAT THE EDGE DOES NOT TEAR OR CUT DIAPHRAGM.

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(7) Tighten diaphragm and retaining plate bolts (13), torque 11 to 15 lb.in., and lockwire.

(8) Locate valve cover (3) on valve housing (22), passing guide pin (4) through center hole of diaphragm retaining plate and diaphragm.

NOTE: Make sure skirt of diaphragm is correctly seated on housing flange as cover is finally positioned on housing studs.

(9) Attach valve cover to valve housing with four washers (9) and self-locking nuts (10). Tighten nuts and torque to 16 to 20 lb.in. Make sure diaphragm still rolls correctly.

(10) After assembly of replacement diaphragm and/or installation of cleaned metering plugs, do adjustment/test.

(11) After tests, retorque cover retaining nuts (10) (Ref. step (9)).

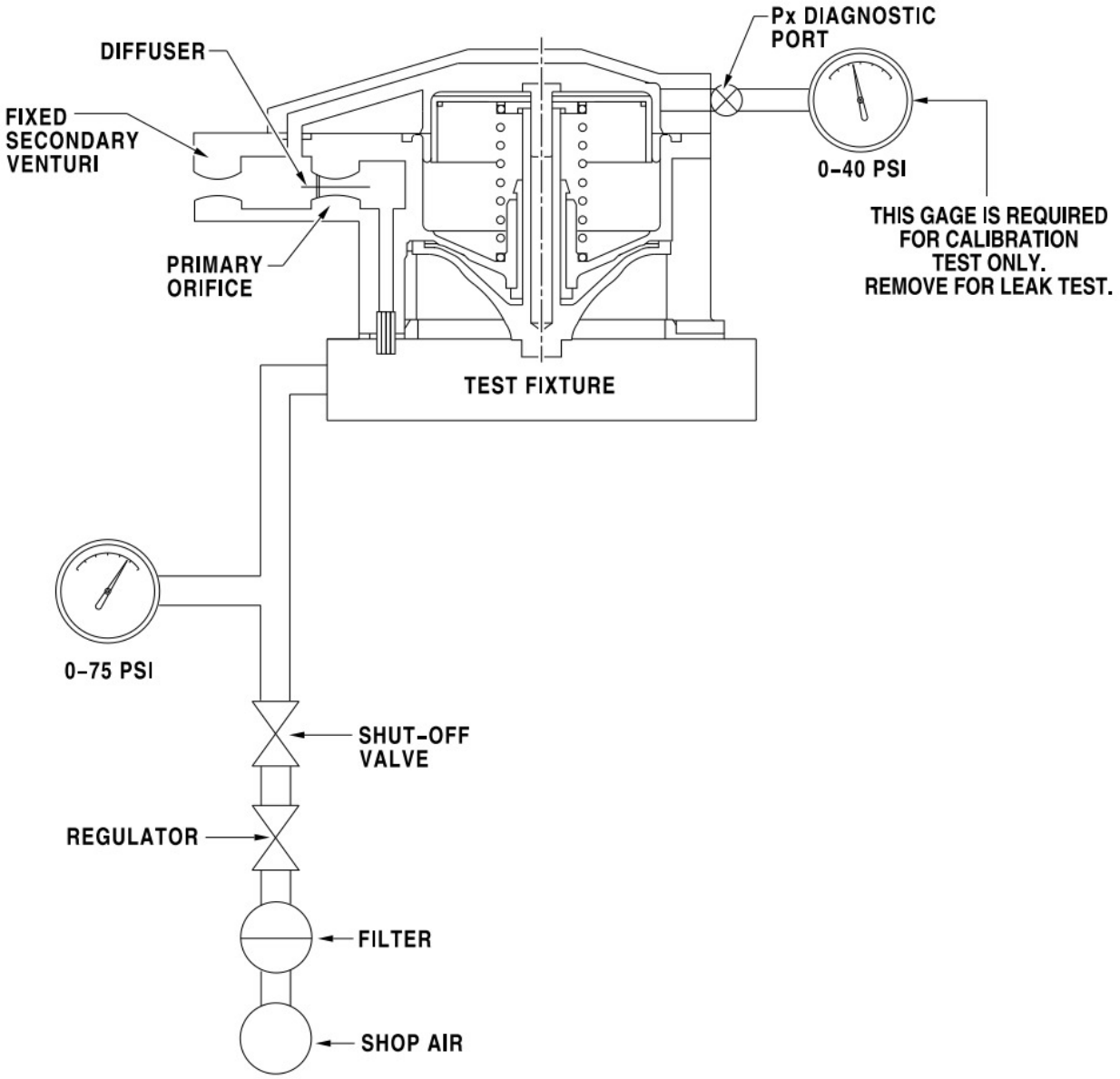
9. Adjustment/Test

CAUTION: OVER PRESSURIZATION OF THE BLEED VALVE BEYOND RECOMMENDED PRESSURE MAY DAMAGE THE BLEED VALVE AND INJURE MAINTENANCE PERSONNEL.

A. Leak Test (Ref. Fig. 201, 203 and 205)

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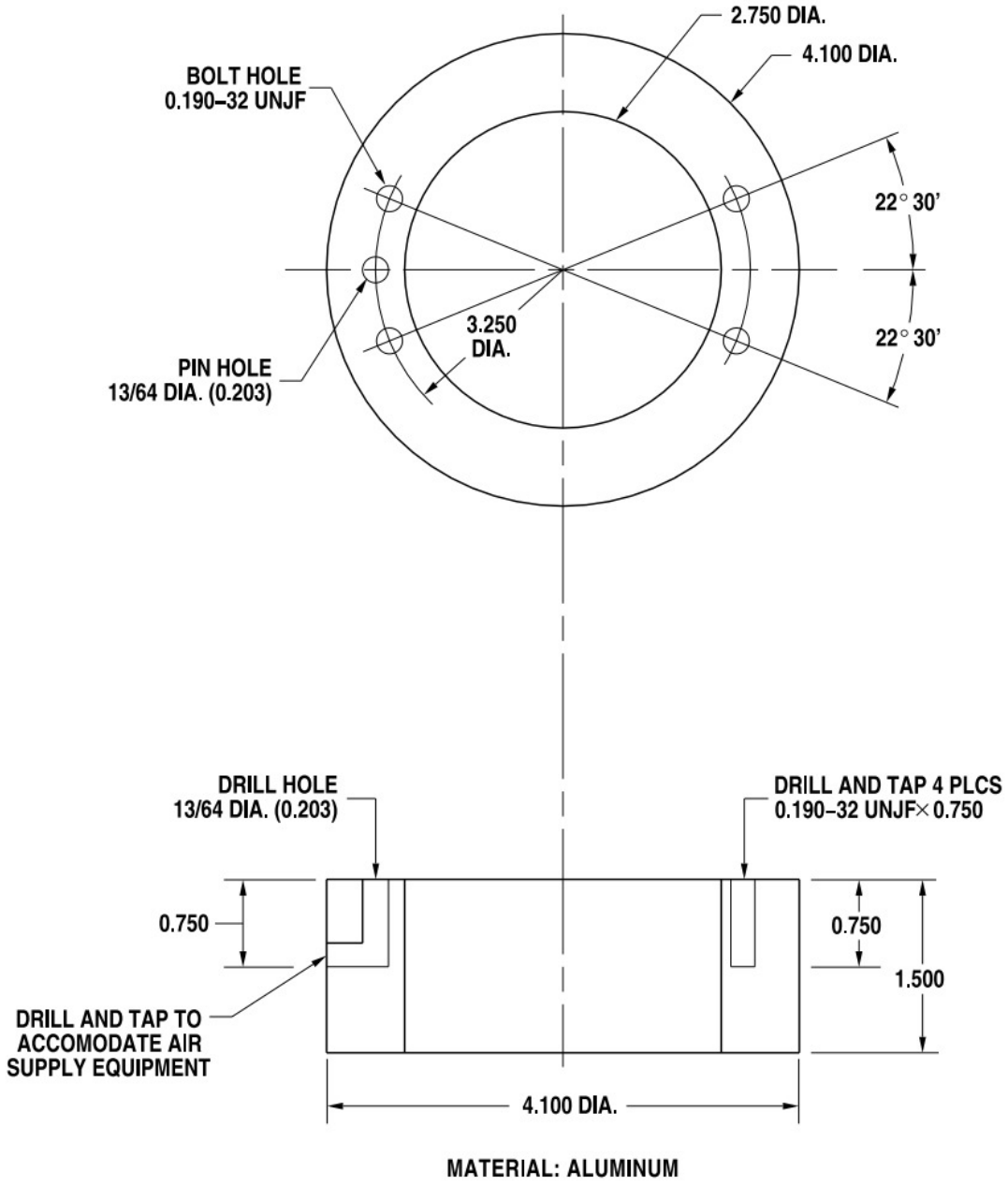
Figure 204 Compressor Bleed Valve Test - Typical



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Figure 205 Compressor Bleed Valve Test Fixture



LOCAL MANUFACTURE

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(1) Install new preformed packing (4, Fig. 201) on spring pin.

(2) Attach bleed valve to the test fixture (Ref. Fig. 205) with bolts and torque 16 to 20 lb.in.

(3) Remove final metering plug (5, Fig. 203) from bleed valve cover and install a suitable blanking plug with preformed packing.

(4) Set the pressure regulator to zero.

(5) Connect a clean dry compressed air supply to fixture.

(6) Open the shut-off valve. Increase the pressure regulator setting until the pressure gage indication is 35 to 45 psig.

(7) Apply leak test fluid (PWC05-007) to all the air line connection, blanking plugs and bleed valve cover flange. Check for leakage. Rectify as required.

(8) With the gage reading 35 to 45 psig, note the reading and close the shut-off valve. A pressure drop over a period of five minutes must not exceed 10 psig.

(9) Disconnect the air supply from regulator. Set pressure regulator to zero to release pressure from bleed valve.

(10) Remove bleed valve from fixture. Remove blanking plug. Install new preformed packing on metering plug and install metering plug in valve cover. Tighten plug and torque 35 to 40 lb.in. Secure with lockwire.

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